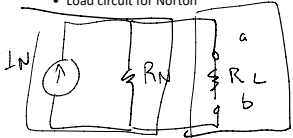
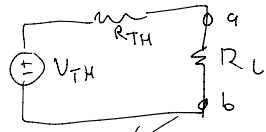


Core principles:

• Load circuit for Norton



• Load circuit for Thevenin



$P_{max} = \frac{V_{TH}^2}{4R_{TH}}$ When $R_L = R_{TH}$

$$P = V \cdot I$$

$$= I^2 \cdot R$$

$$P = \left(\left(\frac{R_N}{R_N + R_L} \right) I_N \right)^2 \cdot R_L$$

$$P(R_L)$$

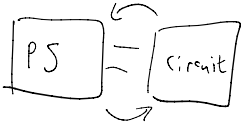
$$P'(R_L) = 0$$

$$R_L = R_{TH} = R_N$$

max Power: $R_{TH} = R_L$

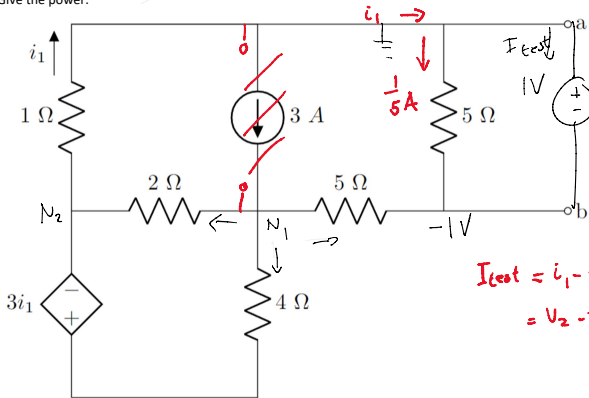
$$P_{max} = \left(\left(\frac{R_N}{2R_N} \right) I_N \right)^2 R_N = \left(\frac{1}{2} I_N \right)^2 R_N = \frac{I_N^2 \cdot R_N}{4}$$

Why do we want to know power/ max power transfer?



Example 1:

Determine the value of R_L that will draw the maximum power from the rest of the circuit. Give the power.



$$R_L = R_{TH} \Rightarrow P_{max}$$

$$(1) R_{TH} = \frac{1}{I_{test}} = R_{load}$$

$$(2) P_{max} = \frac{V_{TH}^2}{4R_{TH}}$$

$$KCL @ N_1: \frac{V_1 - (-1)}{5} + \frac{V_1 - 3i_1 - V_2}{4} + \frac{V_1 - V_2}{2} = 0$$

$$4V_1 + 4 + 5V_1 - 15i_1 - 5V_2 + 10V_1 - 10V_2 = 0$$

$$19V_1 - 15V_2 - 15i_1 = -4$$

$$19V_1 - 30V_2 = -4$$

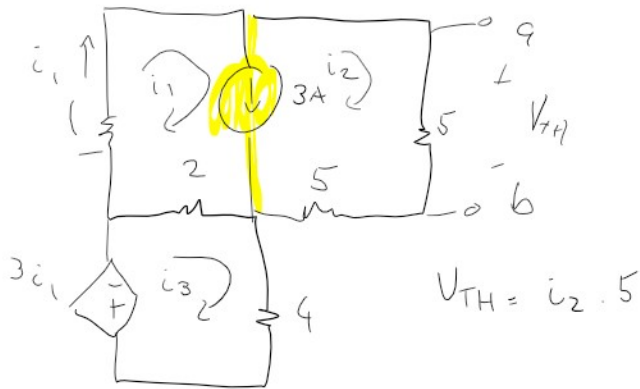
$$V_1:$$

$$V_2:$$

$$KCL @ N_2: \frac{V_2 - V_1}{2} + \frac{V_2 + 3i_1 - V_1}{4} + \frac{V_2}{1} = 0$$

$$2V_2 - 2V_1 + V_2 + 3V_2 - V_1 + 4V_2 = 0$$

$$-3V_1 + 10V_2 = 0$$



$$① \quad i_1 - i_2 = 3$$

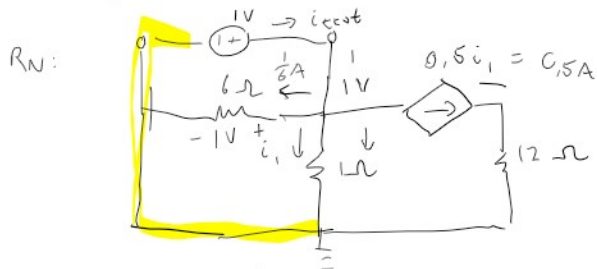
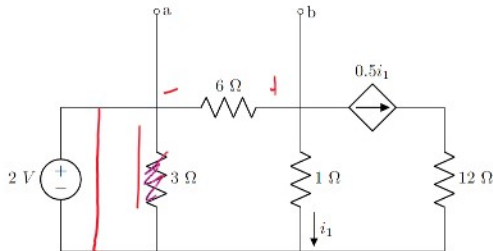
$$② \quad 1 \cdot i_1 + 5i_2 + 5(i_2) + 2(i_1 - i_3) = 0$$

$$3i_1 + 10i_2 - 2i_3 = 0$$

$$③ \quad 4(i_3) + 3i_1 + 2(i_3 - i_1) = 0$$

$$i_1 + 6i_3 = 0$$

To do while getting seated:
Find I_N and R_N between terminals a-b

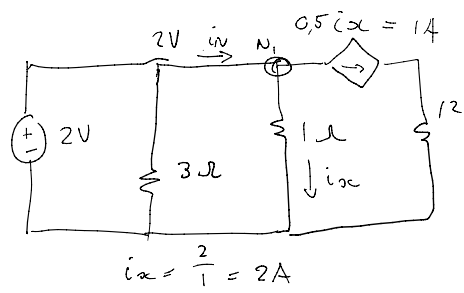


$$i_1 = \frac{1V}{1\Omega} = 1A$$

$$KCL @ Node 1: 0.5A + 1 + \frac{1}{6} - i_{test} = 0$$

$$R_N = \frac{V_{test}}{I_{test}} \quad i_{test} = 1.5 + \frac{1}{6} A$$

$$= \frac{1}{1.5 + \frac{1}{6}} = \Omega$$



$$1A + 2A - I_N = 0$$

$$I_N = 3A$$

